Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe Egli & Sons Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
21810 W August AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 06/01/1954

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X140-X012-XXXX X045-X140-X054-XXXX X045-X170-X002-XXXX X045-X170-X064-XXXX

B. OPERATORS

Egli, Joseph

Operator name: Egli, Joseph

Telephone no.: (209) 634-9559 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
21810 W August AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Egli, Joseph

Legal owner name: Egli, Joseph

Telephone no.: (209) 634-9559 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
21810 W August AVE	Hilmar	CA	95324

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	260	25	59	43	25	57
Maximum number	260	25	59	43	25	57
Average number	43	20	59	43	25	57
Avg live weight (lbs)	1,400	1,500	1,100	800		

Predominant milk cow breed: Holstein

Average milk production: 71 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 2,746.05 *tons per reporting period*

Total nitrogen from manure: 32,007.33 *lbs per reporting period*

After ammonia losses (30% loss applied): 22,405.13 *lbs per reporting period*

Total phosphorus from manure: 4,869.81 *lbs per reporting period*

Total potassium from manure: 8,263.15 *lbs per reporting period*

Total salt from manure: 24,845.55 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,467,200 *gallons*

Total nitrogen generated: 11,167.61 *lbs*

Total phosphorus generated: 1,407.72 *lbs*

Total potassium generated: 10,979.36 *lbs*

Total salt generated: 68,885.43 *lbs*

	12,654,250	*gallons applied*
+	0	*gallons exported*
-	7,187,050	*gallons imported*
=	5,467,200	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - 21536	Ground water
Domestic Well - 21641	Ground water
Domestic Well - 21810	Ground water
Domestic Well - 21889	Ground water
Domestic Well - 9242	Ground water

Source Description	Type
Irrigation Water - Canal	Surface water
Irrigation Water - Well # 1550	Ground water
Irrigation Water - Well # 970	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
07/10/2023	Process wastewater *Lagoon Water Import - Tiberio Azevedo*	336,000.00 *gal*	116.00	25.70	128.00		967
08/12/2023	Process wastewater *Lagoon Water Import - Tiberio Azevedo*	2,142,000.00 *gal*	808.00	77.70	915.00		4,100
08/25/2023	Process wastewater *Lagoon Water Import - Tiberio Azevedo*	2,408,800.00 *gal*	302.00	47.20	622.00		3,500
08/26/2023	Process wastewater *Lagoon Water Import - Tiberio Azevedo*	2,300,250.00 *gal*	237.00	57.10	369.00		2,250

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	25,388.23	3,505.80	36,300.78	189,543.94
Total imports for all materials	25,388.23	3,505.80	36,300.78	189,543.94

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1	32	32	1	process wastewater	X045-X140-X054-XXXX
Field #2	30	30	2	process wastewater	X045-X140-X012-XXXX
Field #3	43	43	1	process wastewater	X045-X140-X012-XXXX
Field #4	26	26	2	process wastewater	X045-X140-X012-XXXX
Field #5	7	7	2	process wastewater	X045-X170-X002-XXXX
Field #6	3	3	2	process wastewater	X045-X170-X002-XXXX
Field #7	14	14	2	process wastewater	X045-X170-X064-XXXX
Totals for areas that were used for application	155	155	12		
Totals for areas that were not used for application					
Land application area totals	155	155	12		

B. CROPS AND HARVESTS

Field #1

Field name: Field #1

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	358.00 *ton*	Dry-weight		70.9	20,300.00	3,900.00	31,200.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.19	132.18	25.39	203.15	579.49

Field #2

Field name: Field #2

Field #2

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	296.00 *ton*	Dry-weight		72.3	21,600.00	4,000.00	31,700.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.87	118.07	21.86	173.28	513.82

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	826.00 *ton*	Dry-weight		70.0	15,300.00	2,700.00	17,700.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.53	252.76	44.60	292.40	892.08

Field #3

Field name: Field #3

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	466.00 *ton*	Dry-weight		71.5	20,400.00	3,700.00	29,600.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.84	126.02	22.86	182.85	494.18

Field #4

Field name: Field #4

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	209.00 *ton*	Dry-weight		72.8	21,000.00	3,700.00	30,600.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.04	91.83	16.18	133.81	345.46

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	570.00 *ton*	Dry-weight		71.0	15,200.00	2,600.00	17,100.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.92	193.27	33.06	217.43	750.21

Field #5

Field name: Field #5

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	56.00 *ton*	Dry-weight		72.8	21,000.00	3,700.00	30,600.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.00	91.39	16.10	133.17	343.81

Field #5

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	141.00 *ton*	Dry-weight		71.0	15,200.00	2,600.00	17,100.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.14	177.58	30.38	199.78	689.29

Field #6

Field name: Field #6

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 3 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	31.00 *ton*	Dry-weight		72.2	20,500.00	3,700.00	30,300.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.33	117.78	21.26	174.08	442.39

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 3 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	79.00 *ton*	Dry-weight		69.3	12,400.00	2,200.00	12,700.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.33	200.49	35.57	205.34	711.42

Field #7

Field name: Field #7

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	133.00 *ton*	Dry-weight		72.2	20,500.00	3,700.00	30,300.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.50	108.28	19.54	160.04	406.71

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	326.00 *ton*	Dry-weight		69.3	12,400.00	2,200.00	12,700.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.29	177.29	31.45	181.58	629.09

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1 - 11/19/2022: Oats, silage-soft dough

Field name: Field #1

Crop: Oats, silage-soft dough — Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	6.19	1.37	10.11	109.53	560,000.00 *gal*
Irrigation Water - Well # 970	Ground water	38.54	0.00	0.00	605.20	2,592,975.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	20.00	2,323,575.00 *gal*
Application event totals		44.74	1.37	10.11	734.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	88.71	8.51	100.22	449.07	420,000.00 *gal*
Irrigation Water - Well # 970	Ground water	37.79	0.00	0.00	593.41	2,542,463.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	16.23	1,885,800.00 *gal*
Application event totals		126.50	8.51	100.22	1,058.70	

Field #2 - 11/19/2022: Oats, silage-soft dough

Field name: Field #2

Crop: Oats, silage-soft dough — Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.07	7.94	69.46	361.53	410,000.00 *gal*
Irrigation Water - Well # 970	Ground water	30.43	0.00	0.00	477.87	1,919,475.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	20.71	2,256,225.00 *gal*
Application event totals		125.50	7.94	69.46	860.12	

Field #2 - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	4.12	1.07	14.32	99.38	440,000.00 *gal*
Irrigation Water - Well # 970	Ground water	29.90	0.00	0.00	469.49	1,885,800.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	20.40	2,222,550.00 *gal*
Application event totals		34.02	1.07	14.32	589.27	

Field #2 - 06/26/2023: Corn, silage

Field name: Field #2

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.37	5,051,250.00 *gal*
Application event totals		0.00	0.00	0.00	46.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	60.86	14.64	94.64	577.06	922,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.97	4,680,825.00 *gal*
Application event totals		60.86	14.64	94.64	620.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.22	8.93	117.65	662.04	680,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.97	4,680,825.00 *gal*
Application event totals		57.22	8.93	117.65	705.00	

Field #2 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	155.36	12.97	113.50	590.80	670,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.80	4,445,100.00 *gal*
Application event totals		155.36	12.97	113.50	631.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	76.74	11.97	157.79	887.91	912,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.73	4,546,125.00 *gal*
Application event totals		76.74	11.97	157.79	929.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.04	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	42.04	

Field #3 - 11/19/2022: Oats, silage-soft dough

Field name: Field #3

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	74.75	7.17	84.45	378.43	475,600.00 *gal*
Irrigation Water - Well # 970	Ground water	27.75	0.00	0.00	435.76	2,508,788.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	22.00	3,434,850.00 *gal*
Application event totals		102.51	7.17	84.45	836.18	

Field #3 - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	27.84	4.34	57.25	322.17	474,300.00 *gal*
Irrigation Water - Well # 970	Ground water	27.94	0.00	0.00	438.68	2,525,625.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	21.78	3,401,175.00 *gal*
Application event totals		55.78	4.34	57.25	782.63	

Field #4 - 11/20/2022: Oats, silage-soft dough

Field name: Field #4

Crop: Oats, silage-soft dough Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.50	7.63	38.00	287.09	925,000.00 *gal*
Irrigation Water - Well # 970	Ground water	27.11	0.00	0.00	425.63	1,481,700.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.25	2,289,900.00 *gal*
Application event totals		61.61	7.63	38.00	736.98	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Well # 970	Ground water	28.65	0.00	0.00	449.82	1,565,888.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	23.90	2,256,225.00 *gal*
Application event totals		28.65	0.00	0.00	473.72	

Field #4 - 05/24/2023: Corn, silage

Field name: Field #4

Crop: Corn, silage Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field #4 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.30	4,276,725.00 *gal*
Application event totals		0.00	0.00	0.00	45.30	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.38	3,434,850.00 *gal*
Application event totals		0.00	0.00	0.00	36.38	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	192.31	18.45	217.26	973.53	739,800.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.02	3,401,175.00 *gal*
Application event totals		192.31	18.45	217.26	1,009.56	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	53.47	12.87	83.14	506.96	702,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.24	3,232,800.00 *gal*
Application event totals		53.47	12.87	83.14	541.20	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.57	1.90	13.99	151.65	630,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.31	3,333,825.00 *gal*
Application event totals		8.57	1.90	13.99	186.97	

Field #4 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.02	3,401,175.00 *gal*
Application event totals		0.00	0.00	0.00	36.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	8.85	1.96	14.44	156.47	650,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.38	3,434,850.00 *gal*
Application event totals		8.85	1.96	14.44	192.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.24	3,232,800.00 *gal*
Application event totals		0.00	0.00	0.00	34.24	

Field #5 - 11/20/2022: Oats, silage-soft dough

Field name: Field #5

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.88	7.31	96.40	542.43	130,000.00 *gal*
Irrigation Water - Well # 970	Ground water	20.59	0.00	0.00	323.37	303,075.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	25.17	639,825.00 *gal*
Application event totals		67.48	7.31	96.40	890.97	

Field #5 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.70	7.15	46.19	281.64	105,000.00 *gal*
Irrigation Water - Well # 970	Ground water	22.88	0.00	0.00	359.30	336,750.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.82	707,175.00 *gal*
Application event totals		52.59	7.15	46.19	668.77	

Field #5 - 05/24/2023: Corn, silage

Field name: Field #5

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.77	909,225.00 *gal*
Application event totals		0.00	0.00	0.00	35.77	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	29.04	6.43	32.04	242.09	210,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.44	875,550.00 *gal*
Application event totals		29.04	6.43	32.04	276.53	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	942,900.00 *gal*
Application event totals		0.00	0.00	0.00	37.09	

Field #5 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.80	808,200.00 *gal*
Application event totals		0.00	0.00	0.00	31.80	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.77	909,225.00 *gal*
Application event totals		0.00	0.00	0.00	35.77	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	165.68	15.90	187.18	838.74	171,600.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.42	976,575.00 *gal*
Application event totals		165.68	15.90	187.18	877.16	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	48.80	11.74	75.88	462.70	172,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.44	875,550.00 *gal*
Application event totals		48.80	11.74	75.88	497.15	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.77	909,225.00 *gal*
Application event totals		0.00	0.00	0.00	35.77	

Field #6 - 11/20/2022: Oats, silage-soft dough

Field name: Field #6

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Field #6 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.79	11.36	149.66	842.15	86,500.00 *gal*
Irrigation Water - Well # 970	Ground water	32.04	0.00	0.00	503.02	202,050.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	21.64	235,725.00 *gal*
Application event totals		104.82	11.36	149.66	1,366.81	

04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	18.03	7.41	61.54	417.25	75,000.00 *gal*
Irrigation Water - Well # 970	Ground water	32.04	0.00	0.00	503.02	202,050.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	21.64	235,725.00 *gal*
Application event totals		50.06	7.41	61.54	941.91	

Field #6 - 05/24/2023: Corn, silage

Field name: Field #6

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.46	538,800.00 *gal*
Application event totals		0.00	0.00	0.00	49.46	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	106.02	16.54	218.00	1,226.72	126,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.37	505,125.00 *gal*
Application event totals		106.02	16.54	218.00	1,273.08	

Field #6 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.82	303,075.00 *gal*
Application event totals		0.00	0.00	0.00	27.82	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	40.66	9.01	44.86	338.92	126,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	404,100.00 *gal*
Application event totals		40.66	9.01	44.86	376.02	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	115.94	9.68	84.70	440.89	50,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.00	370,425.00 *gal*
Application event totals		115.94	9.68	84.70	474.90	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	404,100.00 *gal*
Application event totals		0.00	0.00	0.00	37.09	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	10.82	4.44	36.93	250.35	45,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	404,100.00 *gal*
Application event totals		10.82	4.44	36.93	287.44	

Field #6 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	404,100.00 *gal*
Application event totals		0.00	0.00	0.00	37.09	

Field #7 - 11/20/2022: Oats, silage-soft dough

Field name: Field #7

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.56	10.72	69.28	422.47	315,000.00 *gal*
Irrigation Water - Well # 970	Ground water	31.46	0.00	0.00	494.04	926,063.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.15	1,481,700.00 *gal*
Application event totals		76.02	10.72	69.28	945.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.26	10.89	70.38	429.17	320,000.00 *gal*
Irrigation Water - Well # 970	Ground water	23.45	0.00	0.00	368.28	690,338.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.50	1,347,000.00 *gal*
Application event totals		68.72	10.89	70.38	823.95	

Field #7 - 05/24/2023: Corn, silage

Field name: Field #7

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field #7 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.76	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	37.76	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	55.70	13.40	86.61	528.08	393,750.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.42	1,953,150.00 *gal*
Application event totals		55.70	13.40	86.61	566.50	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	1,885,800.00 *gal*
Application event totals		0.00	0.00	0.00	37.09	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	172.44	16.54	194.82	872.96	357,200.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.43	1,852,125.00 *gal*
Application event totals		172.44	16.54	194.82	909.39	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	7.23	1.88	25.11	174.24	360,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.09	1,885,800.00 *gal*
Application event totals		7.23	1.88	25.11	211.34	

Field #7 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.76	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	37.76	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.44	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	34.44	

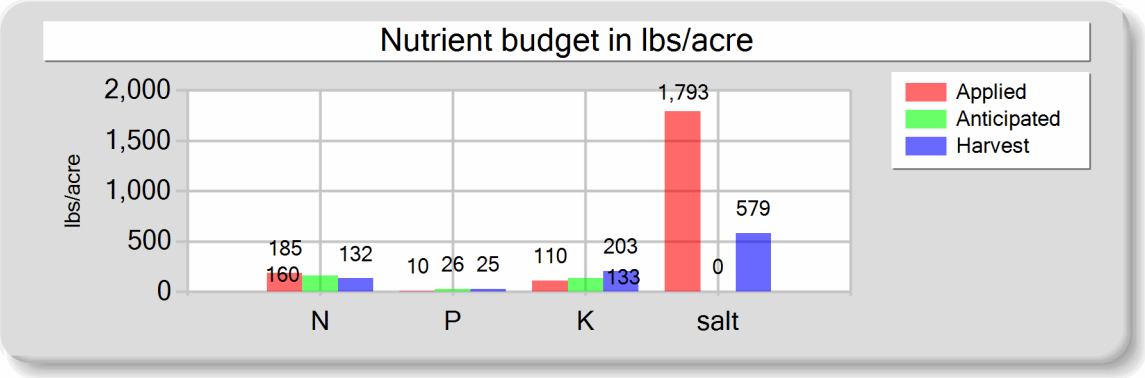
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.76	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	37.76	

B. NUTRIENT BUDGET

Field #1 - 11/19/2022: Oats, silage-soft dough

Field name: Field #1 Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.90	9.88	110.32	558.59
Fresh water	76.34	0.00	0.00	1,234.83
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	185.23	9.88	110.32	1,793.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.18	25.39	203.15	579.49
Nutrient balance	53.06	-15.51	-92.82	1,213.94
Applied to removed ratio	1.40	0.39	0.54	3.09

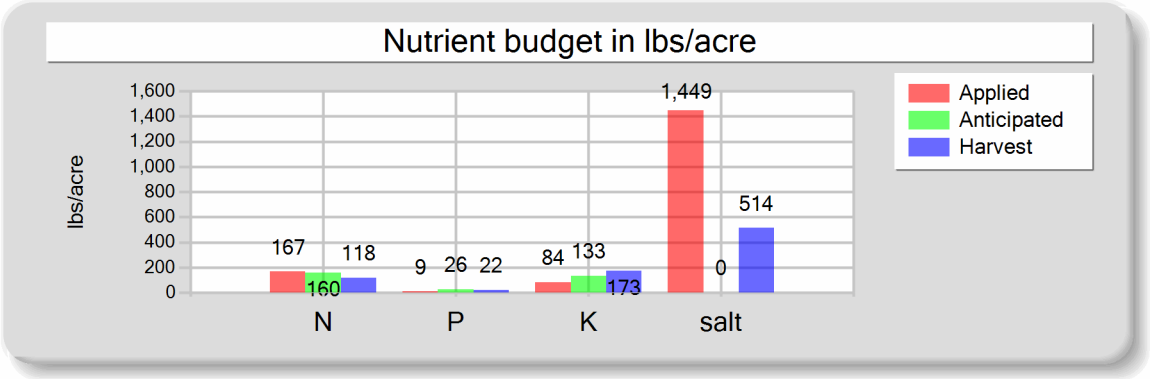
Fresh water applied	
9,344,813.00	*gallons*
344.14	*acre-inches*
10.75	*inches/acre*

Process wastewater applied	
980,000.00	*gallons*
36.09	*acre-inches*
1.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 11/19/2022: Oats, silage-soft dough

Field name: Field #2 Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	99.20	9.01	83.78	460.92
Fresh water	60.33	0.00	0.00	988.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	166.53	9.01	83.78	1,449.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	118.07	21.86	173.28	513.82
Nutrient balance	48.46	-12.85	-89.50	935.57
Applied to removed ratio	1.41	0.41	0.48	2.82

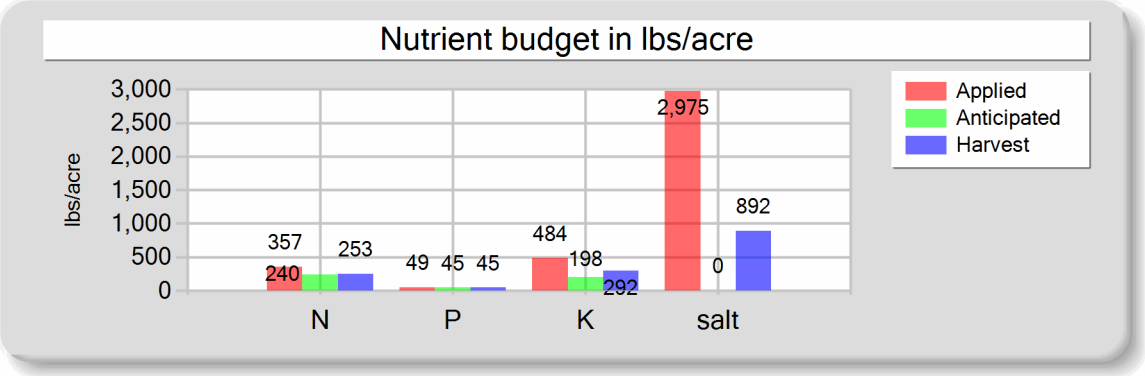
Fresh water applied	
8,284,050.00	*gallons*
305.07	*acre-inches*
10.17	*inches/acre*

Process wastewater applied	
850,000.00	*gallons*
31.30	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 06/26/2023: Corn, silage

Field name: Field #2 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	350.18	48.52	483.58	2,717.80
Fresh water	0.00	0.00	0.00	256.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	357.18	48.52	483.58	2,974.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.76	44.60	292.40	892.08
Nutrient balance	104.42	3.91	191.18	2,082.60
Applied to removed ratio	1.41	1.09	1.65	3.33

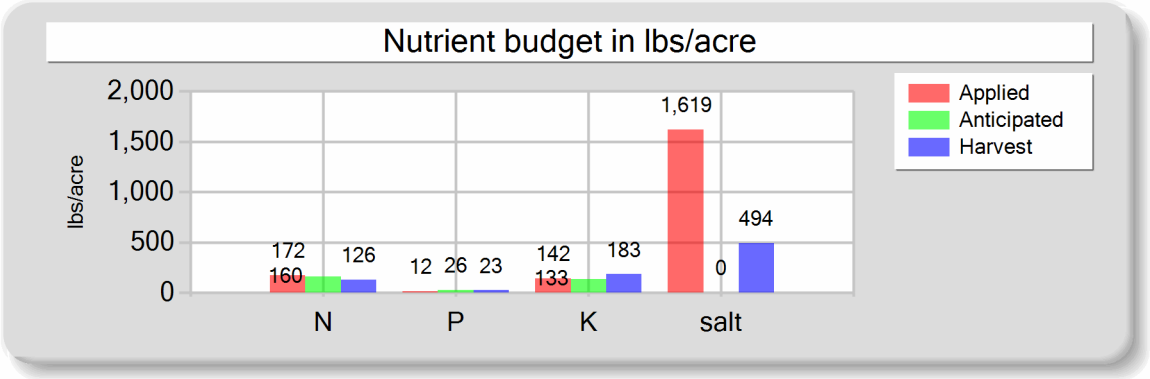
Fresh water applied	
27,983,925.00	*gallons*
1,030.55	*acre-inches*
34.35	*inches/acre*

Process wastewater applied	
3,184,000.00	*gallons*
117.26	*acre-inches*
3.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #3 - 11/19/2022: Oats, silage-soft dough

Field name: Field #3 Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	102.60	11.52	141.71	700.59
Fresh water	55.69	0.00	0.00	918.22
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	172.29	11.52	141.71	1,618.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	126.02	22.86	182.85	494.18
Nutrient balance	46.27	-11.34	-41.14	1,124.64
Applied to removed ratio	1.37	0.50	0.78	3.28

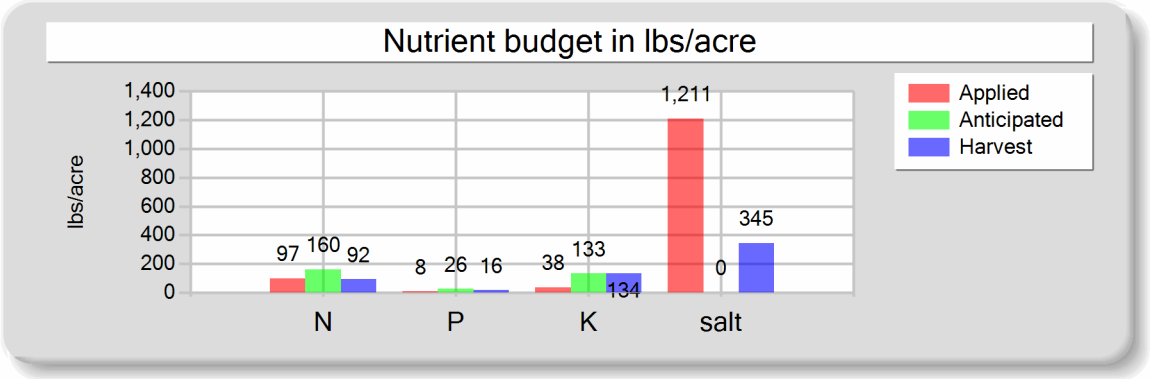
Fresh water applied	
11,870,438.00	*gallons*
437.15	*acre-inches*
10.17	*inches/acre*

Process wastewater applied	
949,900.00	*gallons*
34.98	*acre-inches*
0.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 11/20/2022: Oats, silage-soft dough

Field name: Field #4 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	34.50	7.63	38.00	287.09
Fresh water	55.76	0.00	0.00	923.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	97.25	7.63	38.00	1,210.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	91.83	16.18	133.81	345.46
Nutrient balance	5.42	-8.55	-95.81	865.23
Applied to removed ratio	1.06	0.47	0.28	3.50

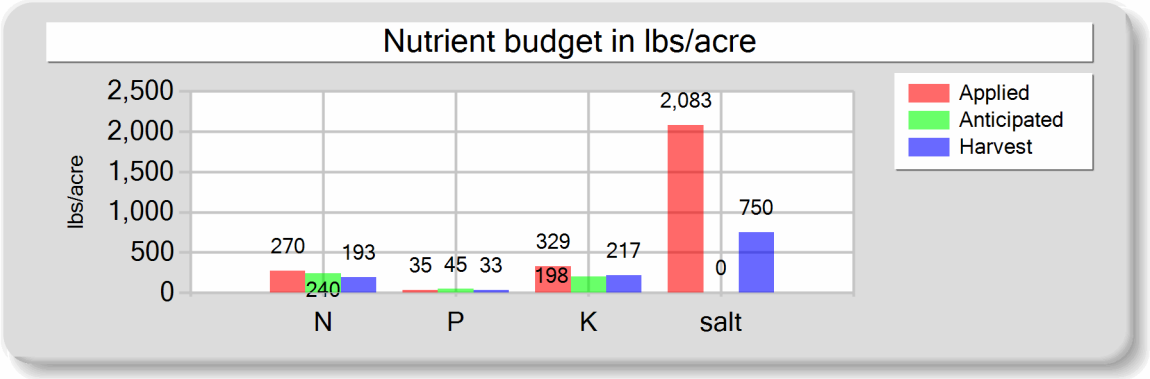
Fresh water applied	
7,593,713.00	*gallons*
279.65	*acre-inches*
10.76	*inches/acre*

Process wastewater applied	
925,000.00	*gallons*
34.06	*acre-inches*
1.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 05/24/2023: Corn, silage

Field name: Field #4 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.20	35.18	328.84	1,788.62
Fresh water	0.00	0.00	0.00	293.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.20	35.18	328.84	2,082.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	193.27	33.06	217.43	750.21
Nutrient balance	76.92	2.12	111.40	1,332.31
Applied to removed ratio	1.40	1.06	1.51	2.78

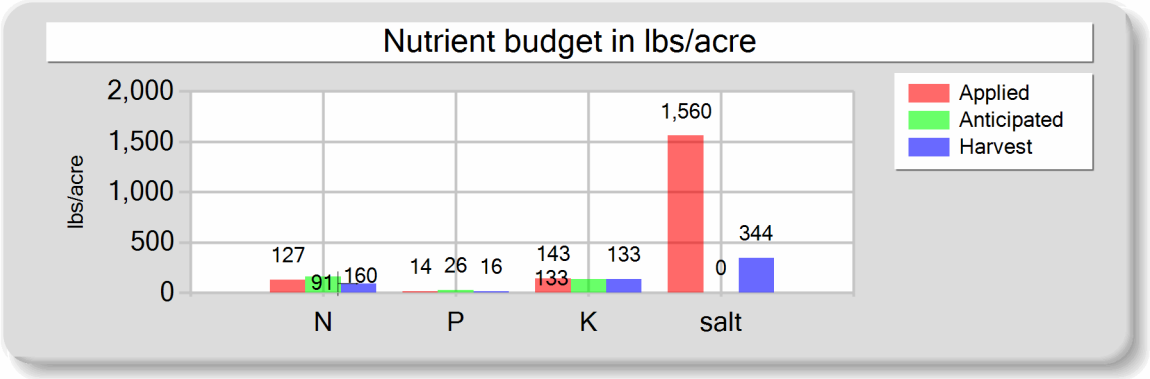
Fresh water applied	
27,748,200.00	*gallons*
1,021.87	*acre-inches*
39.30	*inches/acre*

Process wastewater applied	
2,721,800.00	*gallons*
100.23	*acre-inches*
3.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #5 - 11/20/2022: Oats, silage-soft dough

Field name: Field #5 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	76.59	14.46	142.59	824.07
Fresh water	43.48	0.00	0.00	735.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	127.06	14.46	142.59	1,559.73
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	91.39	16.10	133.17	343.81
Nutrient balance	35.67	-1.64	9.42	1,215.93
Applied to removed ratio	1.39	0.90	1.07	4.54

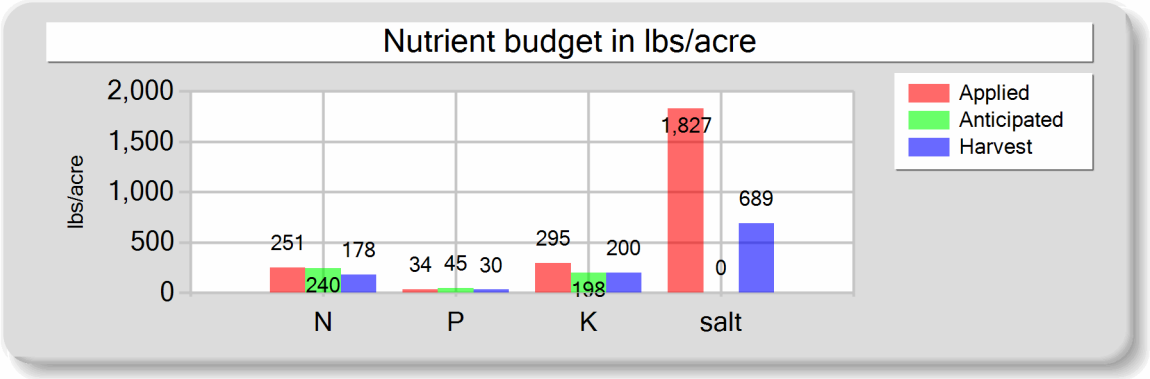
Fresh water applied	
1,986,825.00	*gallons*
73.17	*acre-inches*
10.45	*inches/acre*

Process wastewater applied	
235,000.00	*gallons*
8.65	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #5 - 05/24/2023: Corn, silage

Field name: Field #5 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	243.52	34.07	295.11	1,543.53
Fresh water	0.00	0.00	0.00	283.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.52	34.07	295.11	1,827.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	177.58	30.38	199.78	689.29
Nutrient balance	72.94	3.70	95.33	1,137.75
Applied to removed ratio	1.41	1.12	1.48	2.65

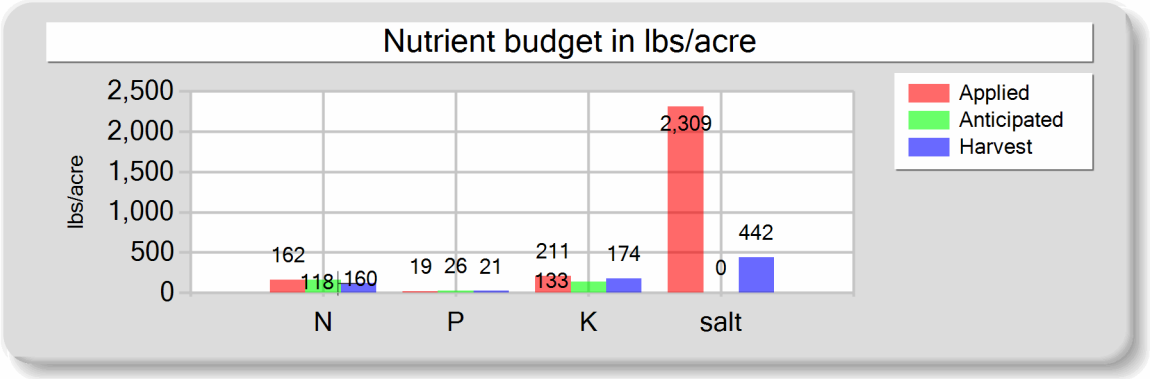
Fresh water applied	
7,206,450.00	*gallons*
265.39	*acre-inches*
37.91	*inches/acre*

Process wastewater applied	
554,100.00	*gallons*
20.41	*acre-inches*
2.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #6 - 11/20/2022: Oats, silage-soft dough

Field name: Field #6 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	90.81	18.76	211.21	1,259.40
Fresh water	64.07	0.00	0.00	1,049.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	161.88	18.76	211.21	2,308.72
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	117.78	21.26	174.08	442.39
Nutrient balance	44.10	-2.49	37.12	1,866.33
Applied to removed ratio	1.37	0.88	1.21	5.22

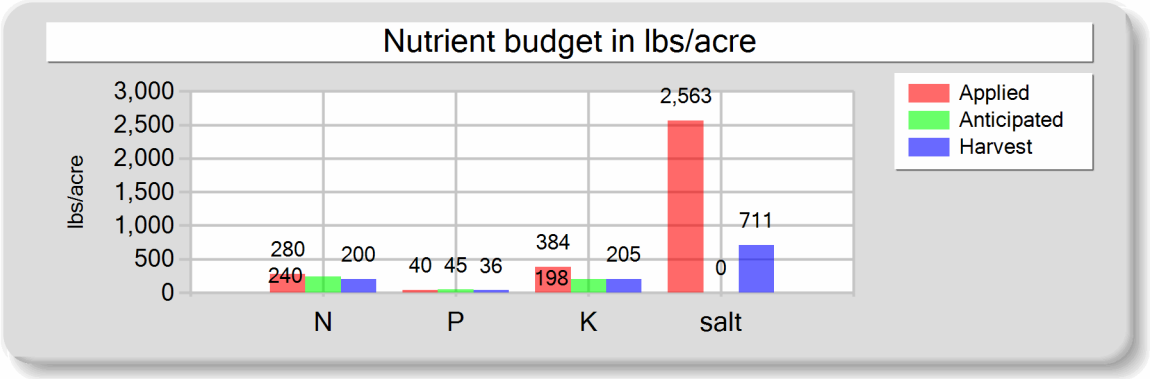
Fresh water applied	
875,550.00	*gallons*
32.24	*acre-inches*
10.75	*inches/acre*

Process wastewater applied	
161,500.00	*gallons*
5.95	*acre-inches*
1.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #6 - 05/24/2023: Corn, silage

Field name: Field #6 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	273.44	39.67	384.50	2,256.88
Fresh water	0.00	0.00	0.00	306.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.44	39.67	384.50	2,562.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	200.49	35.57	205.34	711.42
Nutrient balance	79.94	4.10	179.15	1,851.49
Applied to removed ratio	1.40	1.12	1.87	3.60

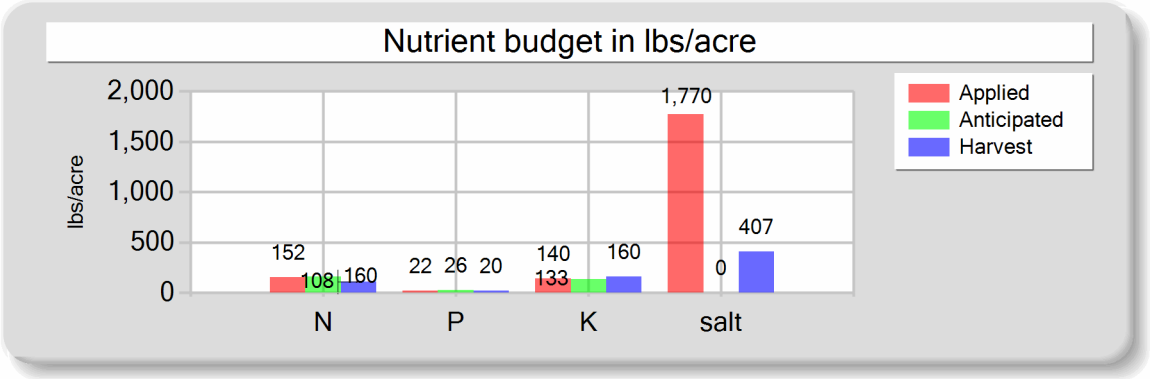
Fresh water applied	
3,333,825.00	*gallons*
122.77	*acre-inches*
40.92	*inches/acre*

Process wastewater applied	
347,000.00	*gallons*
12.78	*acre-inches*
4.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #7 - 11/20/2022: Oats, silage-soft dough

Field name: Field #7 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	89.82	21.61	139.67	851.64
Fresh water	54.92	0.00	0.00	917.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.74	21.61	139.67	1,769.60
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	108.28	19.54	160.04	406.71
Nutrient balance	43.46	2.07	-20.38	1,362.89
Applied to removed ratio	1.40	1.11	0.87	4.35

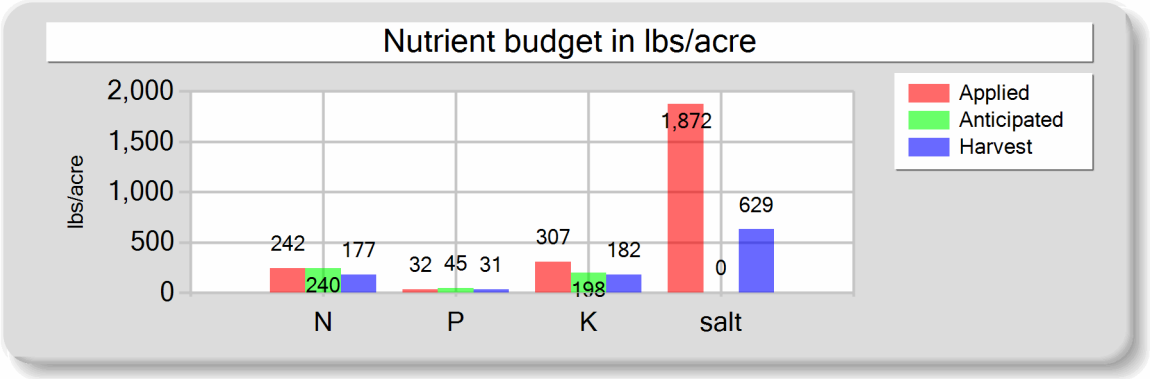
Fresh water applied	
4,445,101.00	*gallons*
163.70	*acre-inches*
11.69	*inches/acre*

Process wastewater applied	
635,000.00	*gallons*
23.38	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #7 - 05/24/2023: Corn, silage

Field name: Field #7 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	235.37	31.83	306.53	1,575.28
Fresh water	0.00	0.00	0.00	296.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.37	31.83	306.53	1,872.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	177.29	31.45	181.58	629.09
Nutrient balance	65.08	0.37	124.95	1,242.95
Applied to removed ratio	1.37	1.01	1.69	2.98

Fresh water applied	
15,086,400.00	*gallons*
555.58	*acre-inches*
39.68	*inches/acre*

Process wastewater applied	
1,110,950.00	*gallons*
40.91	*acre-inches*
2.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	3,900.00	4,300.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	6,500.00	16,300.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	832.00	134.00	0.00	1.60	69.60	609.00								5,180.00	3,170
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water Import - Tiberio Azevedo

Sample and source description: Lagoon Water Import - Tiberio Azevedo

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	808.00	425.00	0.00	1.90	77.70	915.00								9,160.00	4,100
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	38.30	16.10	0.00	4.10	9.40	69.20								1,290.00	750
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water Import - Tiberio Azevedo

Sample and source description: Lagoon Water Import - Tiberio Azevedo

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	116.00	35.50	0.00	0.20	25.70	128.00								1,850.00	967
DL	1.00	0.50	0.10	0.20	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	19.60	2.90	0.00	14.10	8.77	117.00								989.00	812
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

Lagoon Water Import - Tiberio Azevedo

Sample and source description: Lagoon Water Import - Tiberio Azevedo

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	237.00	166.00	0.00	0.30	57.10	369.00								4,040.00	2,250
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	85.90	23.20	0.00	0.50	35.50	295.00								2,090.00	2,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water Import - Tiberio Azevedo

Sample and source description: Lagoon Water Import - Tiberio Azevedo

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	302.00	203.00	0.00	0.00	47.20	622.00								4,790.00	3,500
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - 21536

Domestic Well - 21536

21536

Sample description: 21536

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	40.70	0.00	40.70								1,540.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - 21641

21641

Sample description: 21641

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	41.00	0.00	41.00								1,110.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - 21810

21810

Sample description: 21810

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	50.80	0.00	50.80								1,440.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - 21889

Domestic Well - 21889

21889

Sample description: 21889

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	38.30	0.00	38.30								1,170.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - 9242

9242

Sample description: 9242

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	31.30	0.00	31.30								1,040.00	
DL	0.10	0.10	0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								32.30	33
DL	0.10	0.10	0.10								10.00	10

Irrigation Water - Well # 1550

Irrigation Water - Well # 1550

1550

Sample description: 1550

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	38.90	0.00	38.90								1,050.00	685
DL	0.10	0.10	0.10								10.00	10

Irrigation Water - Well # 970

970

Sample description: 970

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	57.00	0.00	57.00								1,180.00	895
DL	0.10	0.10	0.10								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field #1 - 11/19/2022: Oats, silage-soft dough

Field #1 - 11/19/2022: Oats, silage-soft dough

Field 1

Sample and source description: Field 1

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	3,900.00	31,200.00		8.90
DL	100.00	100.00	100.00		0.01

Field #2 - 11/19/2022: Oats, silage-soft dough

Field 2

Sample and source description: Field 2

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,600.00	4,000.00	31,700.00		9.40
DL	100.00	100.00	100.00		0.01

Field #2 - 06/26/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,700.00	17,700.00		5.40
DL	100.00	100.00	100.00		0.01

Field #3 - 11/19/2022: Oats, silage-soft dough

Field 3

Sample and source description: Field 3

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,400.00	3,700.00	29,600.00		8.00
DL	100.00	100.00	100.00		0.01

Field #4 - 11/20/2022: Oats, silage-soft dough

Field 4 & 5

Sample and source description: Field 4 & 5

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	3,700.00	30,600.00		7.90
DL	100.00	100.00	100.00		0.01

Field #4 - 05/24/2023: Corn, silage

Field 4 & 5

Sample and source description: Field 4 & 5

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,600.00	17,100.00		5.90
DL	100.00	100.00	100.00		0.01

Field #5 - 11/20/2022: Oats, silage-soft dough

Field 4 & 5

Sample and source description: Field 4 & 5

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	3,700.00	30,600.00		7.90
DL	100.00	100.00	100.00		0.01

Field #5 - 05/24/2023: Corn, silage

Field 4 & 5

Sample and source description: Field 4 & 5

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,600.00	17,100.00		5.90
DL	100.00	100.00	100.00		0.01

Field #6 - 11/20/2022: Oats, silage-soft dough

Field 6 & 7

Sample and source description: Field 6 & 7

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,500.00	3,700.00	30,300.00		7.70
DL	100.00	100.00	100.00		0.01

Field #6 - 05/24/2023: Corn, silage

Field 6 & 7

Sample and source description: Field 6 & 7

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,200.00	12,700.00		4.40
DL	100.00	100.00	100.00		0.01

Field #7 - 11/20/2022: Oats, silage-soft dough

Field 6 & 7

Sample and source description: Field 6 & 7

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,500.00	3,700.00	30,300.00		7.70
DL	100.00	100.00	100.00		0.01

Field #7 - 05/24/2023: Corn, silage

Field 6 & 7

Sample and source description: Field 6 & 7

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,200.00	12,700.00		4.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

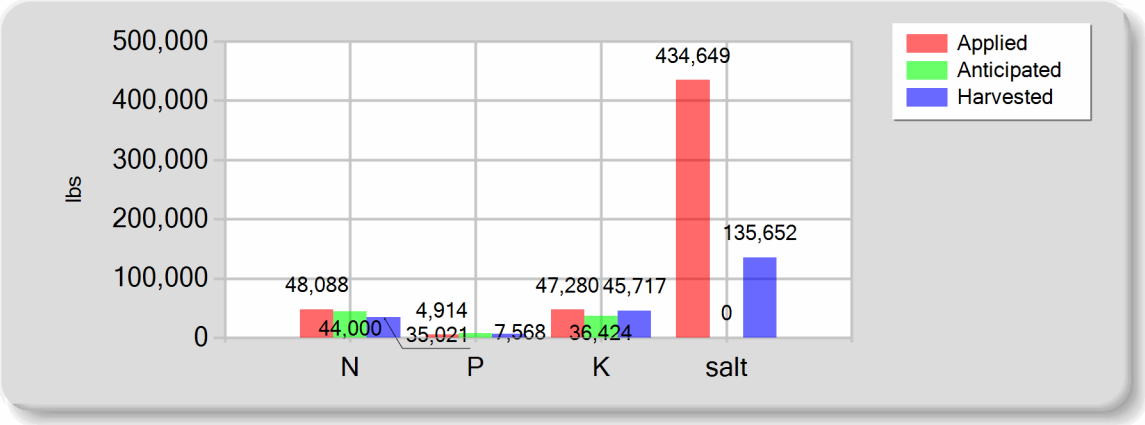
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

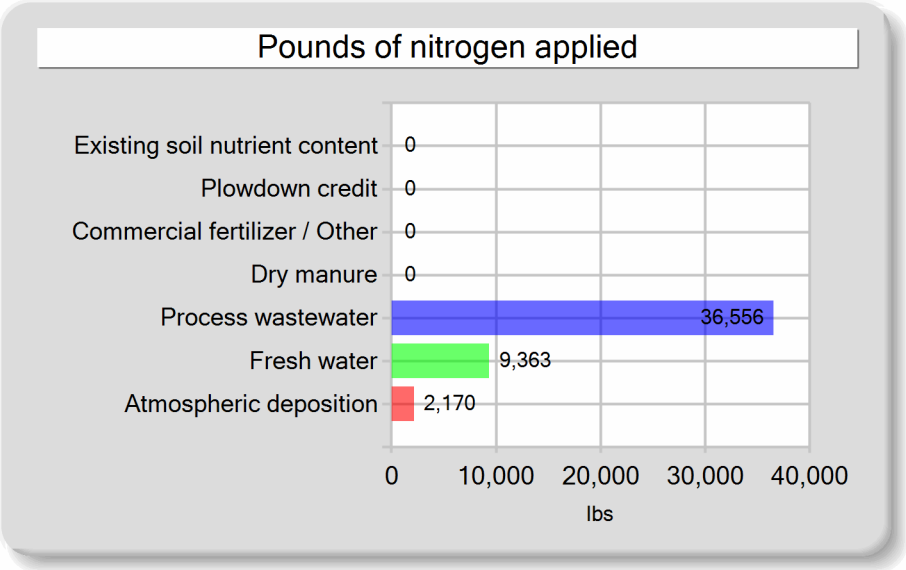
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

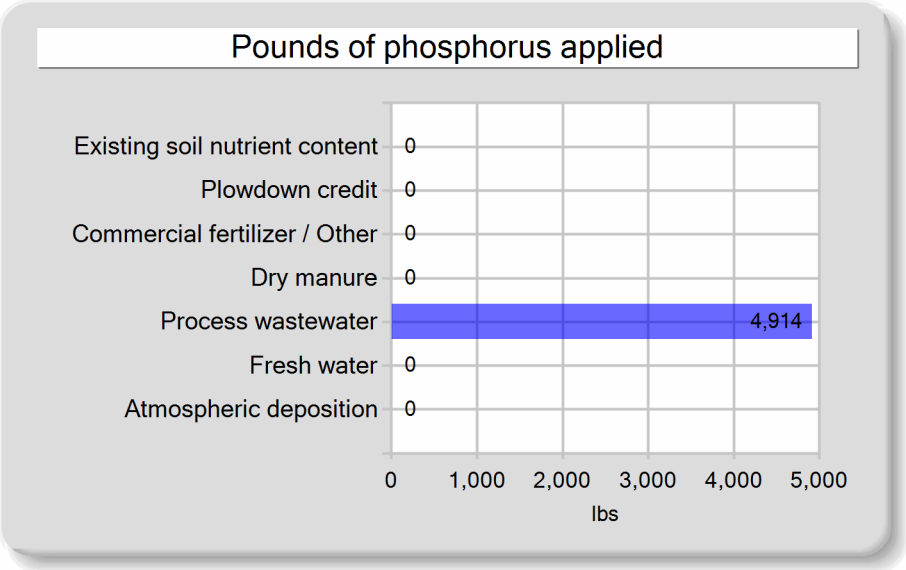
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	36,555.85	4,913.52	47,280.14	258,429.37
Fresh water	9,362.53	0.00	0.00	176,220.01
Atmospheric deposition	2,170.00	0.00	0.00	0.00
Total nutrients applied	48,088.38	4,913.52	47,280.14	434,649.38
Anticipated crop nutrient removal	44,000.00	7,568.00	36,424.00	0.00
Actual crop nutrient removal	35,021.33	6,279.47	45,717.45	135,651.89
Nutrient balance	13,067.06	-1,365.95	1,562.69	298,997.49
Applied to removed ratio	1.37	0.78	1.03	3.20

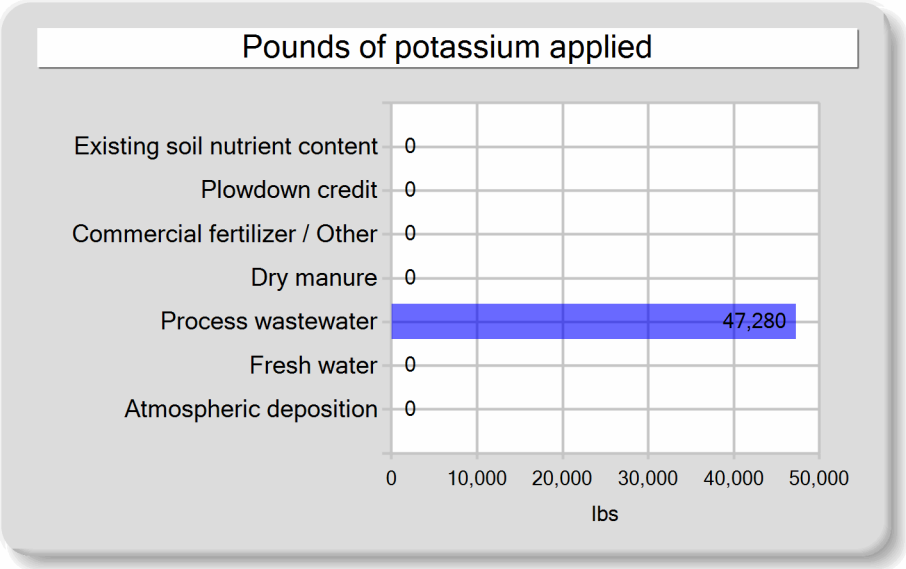
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

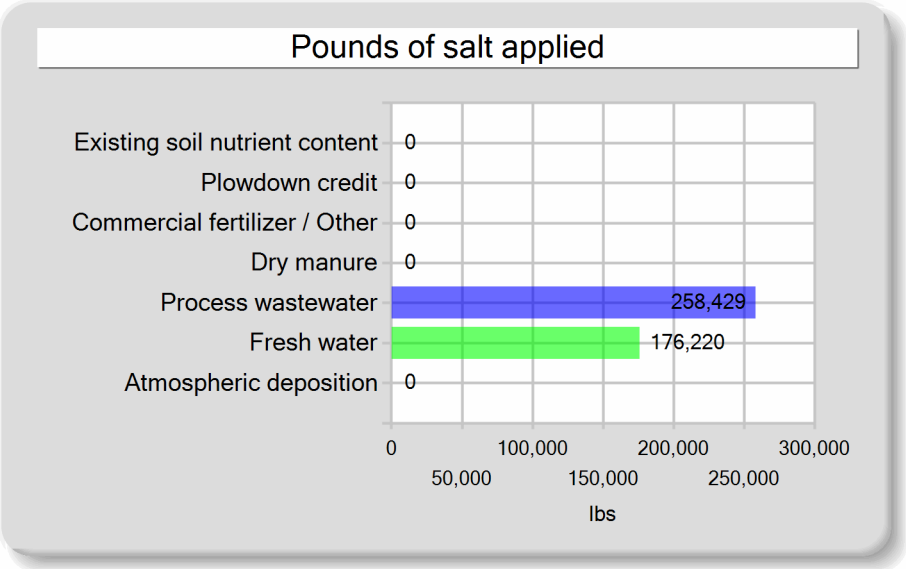


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Milk Cows were SOLD on 3/1/2023 and the facility only raised Heifers from that day forward.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joseph Egli	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-25-2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23G1988-01	970	Ag Water	T. Madruga		07/26/2023 15:30
23G1988-02	1550	Ag Water	T. Madruga		07/26/2023 16:00

Default Cooler Temperature on Receipt °C: 1.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Sample Results

Sample: 970
23G1988-01 (Water)

Sampled: 7/26/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.18**	mmhos/cm	0.01	1		07/31/23 13:04	SM 2510 B		BEG1120
Electrical Conductivity umhos	**1180**	umhos/cm	10.0	1		07/31/23 13:04	SM 2510 B		BEG1120
Nitrate Nitrogen as NO3N	**57.0**	mg/L	0.1	1	10	07/27/23 23:34	EPA 300.0		BEG1055
pH	**7.4**	units	1.0	1		07/31/23 13:04	SM 4500-H+	H	BEG1120
Total Filterable Solids (TDS)	**895**	mg/L	10.0	1		08/03/23 13:15	SM 2540 C		BEG1153
Temperature	**25.0**	°C	0.0	1		07/31/23 13:04	SM 2510 B		BEG1120
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/02/23 09:30	SM 4500-NH3 C		BEH0001
Total Nitrogen	**57.7**	mg/L	1.00	1		08/02/23 09:30	SM 4500-NH3 C		BEH0001

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Sample Results
(Continued)

Sample: 1550
23G1988-02 (Water)

Sampled: 7/26/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.05**	mmhos/cm	0.01	1		07/31/23 13:06	SM 2510 B		BEG1120
Electrical Conductivity umhos	**1050**	umhos/cm	10.0	1		07/31/23 13:06	SM 2510 B		BEG1120
Nitrate Nitrogen as NO3N	**38.9**	mg/L	0.1	1	10	07/27/23 23:54	EPA 300.0		BEG1055
pH	**7.6**	units	1.0	1		07/31/23 13:06	SM 4500-H+	H	BEG1120
Total Filterable Solids (TDS)	**685**	mg/L	10.0	1		08/03/23 13:15	SM 2540 C		BEG1153
Temperature	**25.0**	°C	0.0	1		07/31/23 13:06	SM 2510 B		BEG1120
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/02/23 09:31	SM 4500-NH3 C		BEH0001
Total Nitrogen	**38.9**	mg/L	1.00	1		08/02/23 09:31	SM 4500-NH3 C		BEH0001

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1055										
Blank (BEG1055-BLK1)					Prepared & Analyzed: 7/27/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG1055-BLK2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG1055-BLK3)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEG1055-BS1)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEG1055-BS2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEG1055-DUP1)		**Source: 23G0168-01**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	0.5		0.1	mg/L		0.5			0.197	10
Duplicate (BEG1055-DUP2)		**Source: 23G1989-02**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	23.1		0.1	mg/L		23.2			0.535	10
Matrix Spike (BEG1055-MS1)		**Source: 23G0168-01**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	5.7		0.1	mg/L	5.000	0.5	104	90-110		
Matrix Spike (BEG1055-MS2)		**Source: 23G1989-02**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	28.1		0.1	mg/L	5.000	23.2	98.3	90-110		
Reference (BEG1055-SRM1)					Prepared & Analyzed: 7/27/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEG1055-SRM2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEG1055-SRM3)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1120										
Blank (BEG1120-BLK1)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	5.4		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEG1120-BLK2)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	6.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEG1120-BLK3)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.9		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEG1120-DUP1)		**Source: 23G1984-01**			Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	1.14		0.01	mmhos/cm		1.15			0.750	10
pH	7.3		1.0	units		7.3			0.274	10
Electrical Conductivity umhos	1140		10.0	umhos/cm		1150			0.750	10
Duplicate (BEG1120-DUP2)		**Source: 23G1988-02**			Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	1.05		0.01	mmhos/cm		1.05			0.0475	10
pH	7.6		1.0	units		7.6			0.131	10
Electrical Conductivity umhos	1050		10.0	umhos/cm		1050			0.0475	10
Reference (BEG1120-SRM1)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	527			umhos/cm	538.0		98.0	90-110		
Reference (BEG1120-SRM2)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	7.8			units	7.790		100	.7163-101.28		
Reference (BEG1120-SRM3)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	981			umhos/cm	1000		98.1	90-110		
Electrical Conductivity umhos	981			umhos/cm	1000		98.1	90-110		
Reference (BEG1120-SRM4)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	979			umhos/cm	1000		97.9	90-110		
Electrical Conductivity umhos	979			umhos/cm	1000		97.9	90-110		
Reference (BEG1120-SRM5)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	979			umhos/cm	1000		97.9	90-110		

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Reported: 08/04/2023 11:22

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1120 (Continued)										
Reference (BEG1120-SRM5)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity umhos	979			umhos/cm	1000		97.9	90-110		
Reference (BEG1120-SRM6)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG1120-SRM7)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG1120-SRM8)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1153										
Blank (BEG1153-BLK1)					Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEG1153-BS1)					Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	15.0		10.0	mg/L	2000		0.750	0-200		
Duplicate (BEG1153-DUP1)		**Source: 23G2009-01**			Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	3800		10.0	mg/L		3820			0.656	10
Reference (BEG1153-SRM1)					Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	310			mg/L	325.0		95.4	90-110		

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Ranch: Egli Dairy

Received: 07/27/2023 16:00
Reported: 08/04/2023 11:22

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0001										
Blank (BEH0001-BLK1)					Prepared: 8/1/2023	Analyzed: 8/2/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH0001-BLK2)					Prepared: 8/1/2023	Analyzed: 8/2/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH0001-BS1)					Prepared: 8/1/2023	Analyzed: 8/2/2023				
Kjeldahl Nitrogen (TKN), Total	5.95		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH0001-DUP1)		**Source: 23G1986-01**			Prepared: 8/1/2023	Analyzed: 8/2/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH0001-MS1)		**Source: 23G1986-01**			Prepared: 8/1/2023	Analyzed: 8/2/2023				
Kjeldahl Nitrogen (TKN), Total	7.13		1.40	mg/L	7.992	ND	89.2	90-110		
Reference (BEH0001-SRM1)					Prepared: 8/1/2023	Analyzed: 8/2/2023				
Kjeldahl Nitrogen (TKN), Total	21.8			mg/L	23.80		91.6	90-110		

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07/27/23 16:00 23G1988

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 2 No. of Bottles: 4

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Egli Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

( ) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	970	7/26/2023	1530	1.6	
2	1550	7/26/2023	1600	1.8	
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	7/26/2023 @ 1600	7/27/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	7/27/2023 @ 900	7-27-23 3:35
Third				
Fourth			7/27 1600	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

07/27/23 16:00 23G1988

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice: ☑ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)	1L2	1L2								
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0691-01	9242	Ag Water	T. Madruga		11/13/2023 15:00
23K0691-02	21536	Ag Water	T. Madruga		11/13/2023 15:15
23K0691-03	21641	Ag Water	T. Madruga		11/13/2023 15:30
23K0691-04	21810	Ag Water	T. Madruga		11/13/2023 15:45
23K0691-05	21889	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 4.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Sample Results

Sample: 9242
23K0691-01 (Water)

Sampled: 11/13/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.04**	mmhos/cm	0.01	1		11/17/23 12:22	SM 2510 B		BEK0479
Electrical Conductivity umhos	**1040**	umhos/cm	10.0	1		11/17/23 12:22	SM 2510 B		BEK0479
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0437
Nitrate Nitrogen as NO3N	**31.3**	mg/L	0.1	1	10	11/15/23 03:23	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 12:22	SM 4500-H+	H	BEK0479
pH	**7.8**	units	1.0	1		11/17/23 12:22	SM 4500-H+	H	BEK0479

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Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Sample Results
(Continued)

Sample: 21536
23K0691-02 (Water)

Sampled: 11/13/2023 15:15
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		11/17/23 12:23	SM 2510 B		BEK0479
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		11/17/23 12:23	SM 2510 B		BEK0479
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:15	Field		BEK0437
Nitrate Nitrogen as NO3N	**40.7**	mg/L	0.1	1	10	11/15/23 03:42	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 12:23	SM 4500-H+	H	BEK0479
pH	**7.9**	units	1.0	1		11/17/23 12:23	SM 4500-H+	H	BEK0479

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Account Manager: Ben Nydam
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Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Sample Results
(Continued)

Sample: 21641
23K0691-03 (Water)

Sampled: 11/13/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.11**	mmhos/cm	0.01	1		11/17/23 12:25	SM 2510 B		BEK0479
Electrical Conductivity umhos	**1110**	umhos/cm	10.0	1		11/17/23 12:25	SM 2510 B		BEK0479
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:30	Field		BEK0437
Nitrate Nitrogen as NO3N	**41.0**	mg/L	0.1	1	10	11/15/23 04:02	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 12:25	SM 4500-H+	H	BEK0479
pH	**7.8**	units	1.0	1		11/17/23 12:25	SM 4500-H+	H	BEK0479

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Account Manager: Ben Nydam
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Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Sample Results
(Continued)

Sample: 21810
23K0691-04 (Water)

Sampled: 11/13/2023 15:45
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.44**	mmhos/cm	0.01	1		11/17/23 12:26	SM 2510 B		BEK0479
Electrical Conductivity umhos	**1440**	umhos/cm	10.0	1		11/17/23 12:26	SM 2510 B		BEK0479
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:45	Field		BEK0437
Nitrate Nitrogen as NO3N	**50.8**	mg/L	0.1	1	10	11/15/23 04:22	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 12:26	SM 4500-H+	H	BEK0479
pH	**7.6**	units	1.0	1		11/17/23 12:26	SM 4500-H+	H	BEK0479

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Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Sample Results
(Continued)

Sample: 21889
23K0691-05 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.17**	mmhos/cm	0.01	1		11/17/23 12:28	SM 2510 B		BEK0479
Electrical Conductivity umhos	**1170**	umhos/cm	10.0	1		11/17/23 12:28	SM 2510 B		BEK0479
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0437
Nitrate Nitrogen as NO3N	**38.3**	mg/L	0.1	1	10	11/15/23 04:41	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 12:28	SM 4500-H+	H	BEK0479
pH	**7.9**	units	1.0	1		11/17/23 12:28	SM 4500-H+	H	BEK0479

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0466										
Blank (BEK0466-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0466-BS1)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEK0466-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEK0466-DUP1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	20.3		0.1	mg/L		20.3			0.133	10
Duplicate (BEK0466-DUP2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	31.0		0.1	mg/L		31.3			1.08	10
Matrix Spike (BEK0466-MS1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	25.8		0.1	mg/L	5.000	20.3	110	90-110		
Matrix Spike (BEK0466-MS2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.1		0.1	mg/L	5.000	31.3	96.8	90-110		
Reference (BEK0466-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0479										
Blank (BEK0479-BLK1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0479-BLK2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEK0479-BLK3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.0		1.0	units						
Duplicate (BEK0479-DUP1)		**Source: 23K0693-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1.52		0.01	mmhos/cm		1.49			1.68	10
pH	8.0		1.0	units		8.0			0.501	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1490			1.68	10
Duplicate (BEK0479-DUP2)		**Source: 23K0706-01**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	0.94		0.01	mmhos/cm		0.93			0.428	10
Electrical Conductivity umhos	936		10.0	umhos/cm		932			0.428	10
pH	7.2		1.0	units		7.1			0.559	10
Reference (BEK0479-SRM1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	573			umhos/cm	538.0		107	90-110		
Reference (BEK0479-SRM2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0479-SRM3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0479-SRM4)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0479-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Egli Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:55

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0479 (Continued)										
Reference (BEK0479-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEK0479-SRM6)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0479-SRM7)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0479-SRM8)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/14/23 15:45 23K0691

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 5 No. of Bottles: 5

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Egli Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	9242	11/13/2023	1500	4.5	ND
2	21536	11/13/2023	1515	5.6	ND
3	21641	11/13/2023	1530	4.2	ND
4	21810	11/13/2023	1545	1.4	ND
5	21889	11/13/2023	1600	1.9	ND
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:30
Third				
Fourth		DLI	11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0691

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1	1	1					
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										